# OSSEC (Ubuntu)

## Aim

Install and configure **OSSEC** and demonstrate:

- Log Monitoring
- File Integrity Monitoring (FIM)
- Rootkit Detection
- Active Response
- Alert Generation

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# Lab Setup

```
Machine 1:
Ubuntu Server
IP: 192.168.1.20

(Optional)
Machine 2:
Windows/Linux Client
IP: 192.168.1.30
```

For CDAC lab, a **single Ubuntu machine** is enough.

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# Step 1: Update Ubuntu

```
sudo apt update
sudo apt upgrade -y
```

---

## Step 2: Install Required Packages

```
sudo apt install -y \
build-essential \
gcc \
g++ \
make \
git \
wget \
unzip
```

Verify:

```
gcc --version
```

---

## Step 3: Download OSSEC

```
cd /tmp

wget https://github.com/ossec/ossec-hids/archive/master.zip

unzip master.zip

cd ossec-hids-master
```

Verify:

```
ls
```

You should see:

```
install.sh
src
```

```
etc
active-response
```

---

# Step 4: Start Installation

```
sudo ./install.sh
```

---

# Step 5: Installation Wizard

### Language

```
Language?
```

Press:

```
ENTER
```

---

### Installation Type

Choose:

```
local
```

Example:

```
1- What kind of installation?

local
```

For single-machine lab.

---

## Installation Directory

Default:

```
/var/ossec
```

Press:

```
ENTER
```

---

## Email Notification

```
Do you want e-mail notification?
```

Choose:

```
n
```

---

## Integrity Check Daemon

```
Enable integrity check daemon?
```

Choose:

```
y
```

This enables File Integrity Monitoring.

---

## Rootkit Detection

```
Enable rootkit detection?
```

Choose:

```
y
```

---

### Active Response

```
Enable active response?
```

Choose:

```
y
```

---

Installation completes.

Example:

```
OSSEC HIDS finished successfully.
```

---

# Step 6: Start OSSEC

```
sudo /var/ossec/bin/ossec-control start
```

Check:

```
sudo /var/ossec/bin/ossec-control status
```

Expected:

```
ossec-analysisd running
ossec-logcollector running
ossec-syscheckd running
ossec-execd running
```

---

# Step 7: Understand Important Directories

### Configuration

```
/var/ossec/etc/ossec.conf
```

### Alerts

```
/var/ossec/logs/alerts/alerts.log
```

### Rules

```
/var/ossec/rules
```

### Logs

```
/var/ossec/logs
```

---

# Step 8: Monitor Authentication Logs

Open configuration:

```
sudo nano /var/ossec/etc/ossec.conf
```

Find:

```
<localfile>
```

Add:

```
<localfile>
  <log_format>syslog</log_format>
```

```
  <location>/var/log/auth.log</location>
</localfile>
```

Save.

Restart:

```
sudo /var/ossec/bin/ossec-control restart
```

---

# Step 9: Generate Failed Login Alert

Open another terminal.

Attempt:

```
ssh fakeuser@localhost
```

Enter wrong password several times.

---

# Step 10: View Alerts

```
sudo tail -f /var/ossec/logs/alerts/alerts.log
```

Expected:

```
Authentication failed
Invalid user
Multiple login failures
```

OSSEC has detected the attack.

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# Step 11: File Integrity Monitoring (FIM)

Open configuration:

```
sudo nano /var/ossec/etc/ossec.conf
```

Find:

```
<syscheck>
```

Add:

```
<directories check_all="yes">
/etc
</directories>
```

Save.

Restart:

```
sudo /var/ossec/bin/ossec-control restart
```

---

# Step 12: Generate File Change Alert

Create a test file:

```
sudo touch /etc/testfile
```

Wait a few minutes.

Check alerts:

```
sudo tail -f /var/ossec/logs/alerts/alerts.log
```

Expected:

```
File added:
/etc/testfile
```

---

## Step 13: Modify Existing File

```
sudo echo "test" >> /etc/hosts
```

Alert:

```
File modified:
/etc/hosts
```

---

## Step 14: Rootkit Detection

Run:

```
sudo /var/ossec/bin/rootcheck_control restart
```

Check logs:

```
cat /var/ossec/logs/rootcheck/rootcheck.log
```

Example:

```
Rootkit scan completed
```

---

## Step 15: Active Response

OSSEC can automatically block attackers.

Edit:

```
sudo nano /var/ossec/etc/ossec.conf
```

Add:

```
<active-response>
  <command>host-deny</command>
  <location>local</location>
  <level>10</level>
</active-response>
```

Restart:

```
sudo /var/ossec/bin/ossec-control restart
```

Now severe attacks can trigger automatic blocking.

---

# Step 16: Check OSSEC Logs

```
sudo tail -f /var/ossec/logs/ossec.log
```

Example:

```
Analysis daemon started
Syscheck started
Rootcheck started
```

---

# Step 17: Useful Commands

### Start

```
sudo /var/ossec/bin/ossec-control start
```

### Stop

```
sudo /var/ossec/bin/ossec-control stop
```

### Restart

```
sudo /var/ossec/bin/ossec-control restart
```

### Status

```
sudo /var/ossec/bin/ossec-control status
```

---

# Architecture

```
      System Logs



     OSSEC Log Collector



     Analysis Engine




Log      File Integrity  Rootkit
Analysis Monitoring      Detection




       Alerts
```

---

# Expected Lab Demonstration

## Demo 1

```
ssh fakeuser@localhost
```

Result:

```
Authentication Failure Alert
```

---

## Demo 2

```
touch /etc/testfile
```

Result:

```
File Added Alert
```

---

## Demo 3

```
echo test >> /etc/hosts
```

Result:

```
File Modified Alert
```

---

## What is OSSEC?

Host-Based Intrusion Detection System (HIDS).

## What is FIM?

File Integrity Monitoring.

## Alert File?

```
/var/ossec/logs/alerts/alerts.log
```

## Configuration File?

```
/var/ossec/etc/ossec.conf
```

## Difference between OSSEC and Snort?

| OSSEC | Snort |
|---|---|
| HIDS | NIDS |
| Host Monitoring | Network Monitoring |
| File Integrity | Packet Inspection |

**OSSEC is an open-source HIDS that performs log analysis, file integrity monitoring, rootkit detection, and active response to security threats.**